



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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August 29 2025



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
 - Data collection using SpaceX API
 - Web scrapping on wikipedia webpage
 - Data wrangling
 - Exploratory Data Analysis using SQL and charts
 - Spatial visualization using folium
 - Dashboard for interactive data exploration
- Summary of all results
 - Machine learning training and prediction
 - Model uses important features to predict landing success such as payload mass, launching site and orbit type

Introduction

Project background and context

- Space X advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars; other providers cost upward of 165 million dollars each, much of the savings is because Space X can reuse the first stage. Therefore, if we can determine if the first stage will land, we can determine the cost of a launch. This information can be used if an alternate company wants to bid against space X for a rocket launch. This goal of the project is to create a machine learning pipeline to predict if the first stage will land successfully.

Problems you want to find answers

- What factors determine if the rocket will land successfully?
- The interaction amongst various features that determine the success rate of a successful landing.
- What operating conditions needs to be in place to ensure a successful landing program.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Data collection using SpaceX API
 - Web scrapping on wikipedia webpage
- Perform data wrangling
 - Handling missing information
 - New variable: successful landing
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models using sklearn
 - Splitting the data between training and testing dataset
 - Modeling and finding the best parameters for each method tested
 - Confusion matrix to show the model accuracy

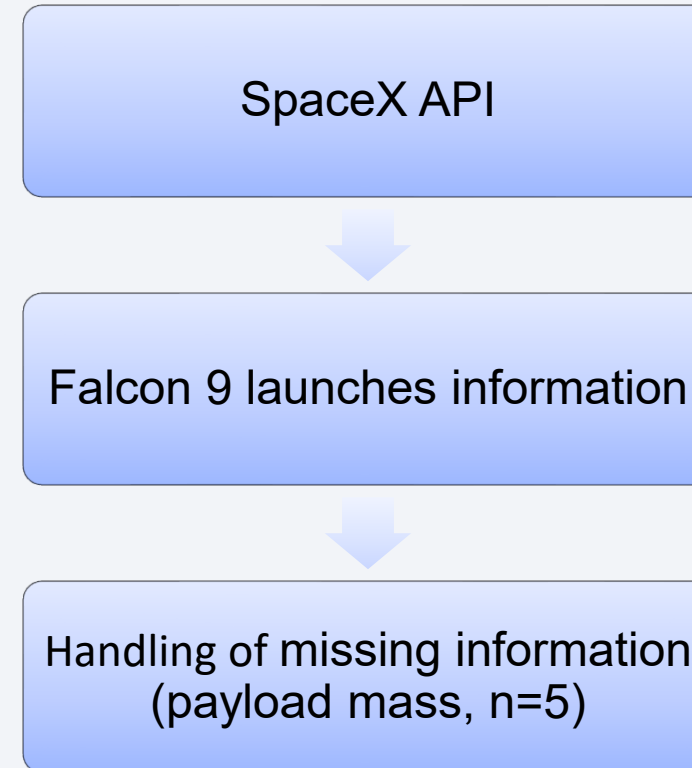
Data Collection

Data was collected on the SpaceX REST API which gives us a lot of information of their rocket launches.

Another source of information was found on Wikipedia, where we scraped data from their website.

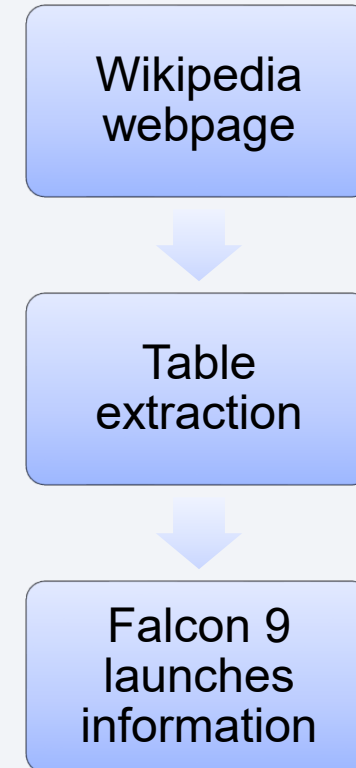
Data Collection – SpaceX API

- We requested rocket launch data from SpaceX API, using this website: <https://api.spacexdata.com/v4/launches/past>
- We selected only the data we were interested (17 columns – features) for Falcon9 launches (90 launches in total)
- We handled missing information (Payload Mass on 5 launches)



Data Collection - Scraping

- We also scrapped data from the Wikipedia page titled:
https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922
(version of 9th June 2021)
- We extracted the data from all the tables of the webpage



Data Wrangling

- Exploratory data analysis to find some patterns in the data
- Creation of a landing outcome label (0: bad landing, 1 good landing)

EDA with Data Visualization

To understand the data, we created a series of charts:

- Scatter plot of Flight Number vs Payload Mass (showing the successful or failed landings)
- Scatter plot of Flight Number vs Launch Sites (showing the successful or failed landings)
- Scatter plot of Payload Mass vs Launch Sites (showing the successful or failed landings)
- Barplot of Orbit type vs Success rate
- Scatter plot of Flight Number vs Orbit type (showing the successful or failed landings)
- Scatter plot of Payload Mass vs Orbit type (showing the successful or failed landings)
- Line plot of year vs Success rate

EDA with Data SQL

The following SQL queries were performed:

- Names of the unique launch sites in the space mission;
- First 5 launch sites whose name begins with the string 'CCA';
- Sum of the payload mass carried by boosters launched by NASA (CRS);
- Average payload mass carried by booster version F9 v1.1;
- Date when the first successful landing outcome in ground pad was achieved;
- Names of the boosters which have success in drone ship and have payload mass between 4000 and 6000 kg;
- Total number of successful and failure mission outcomes;
- Names of the booster versions which have carried the maximum payload mass (15600kg);
- Failed landing out comes in droneship, their booster versions, and launch site names for in year 2015; and
- Rank of the count of landing outcomes (such as Failure (droneship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20.

Build an Interactive Map with Folium

Creation of folium maps to highlight:

- Markers to show launch sites, as well as lines to indicate distances between the launch sites and the coast, the closest railway, the closest city, etc.

Build a Dashboard with Plotly Dash

We built a dashboard to interactively show important information, especially information on launches for each launch site, and the relationship between payload mass and the outcome of the landing for the different booster versions

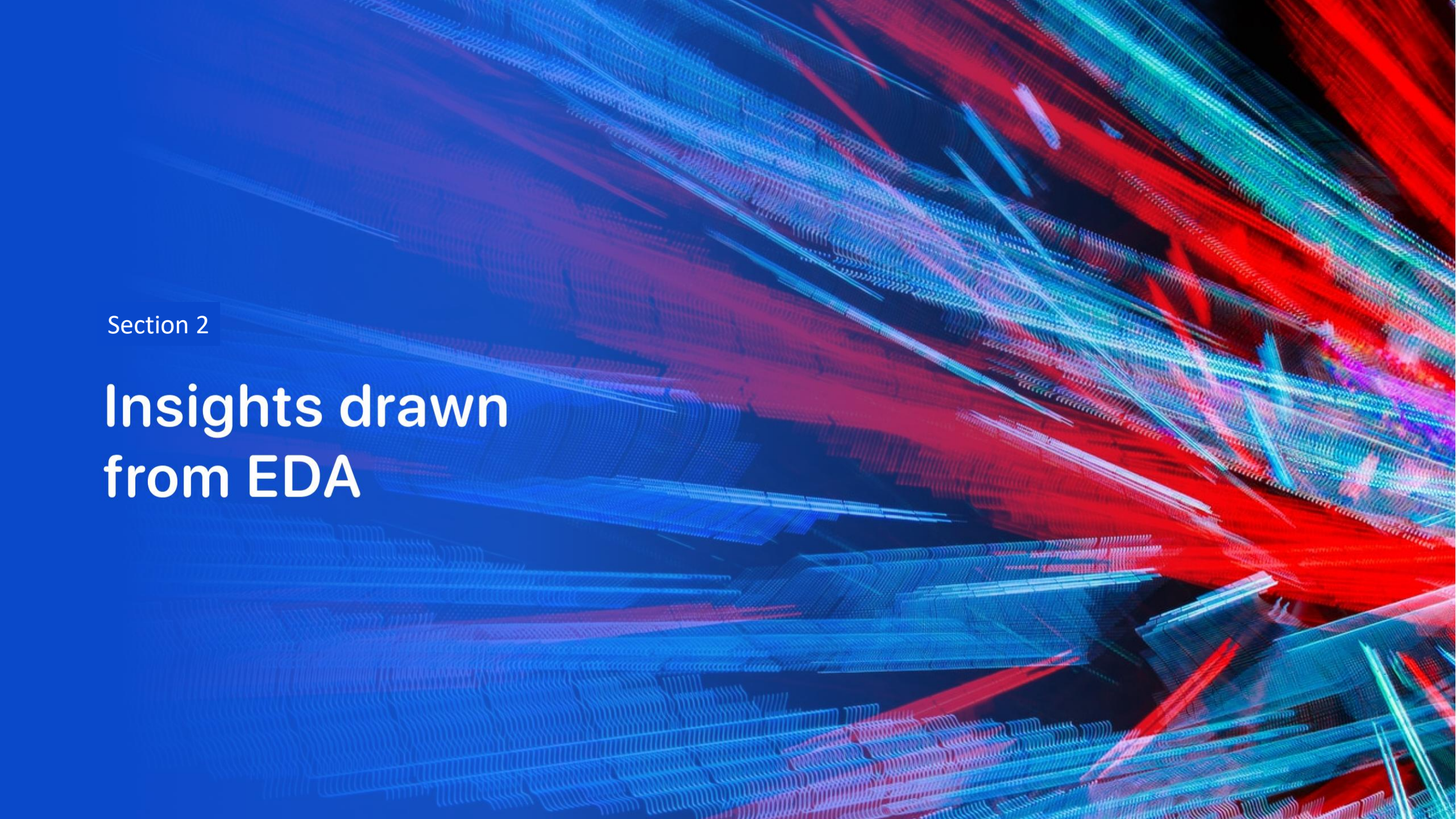
Predictive Analysis (Classification)

- We transformed the data, split our data into training (70%) and testing (30%).
- We built different machine learning models to predict the landing outcome and tune different hyperparameters using a grid search method .
- We selected the best performing classification model.

Results

Exploratory data analysis results:

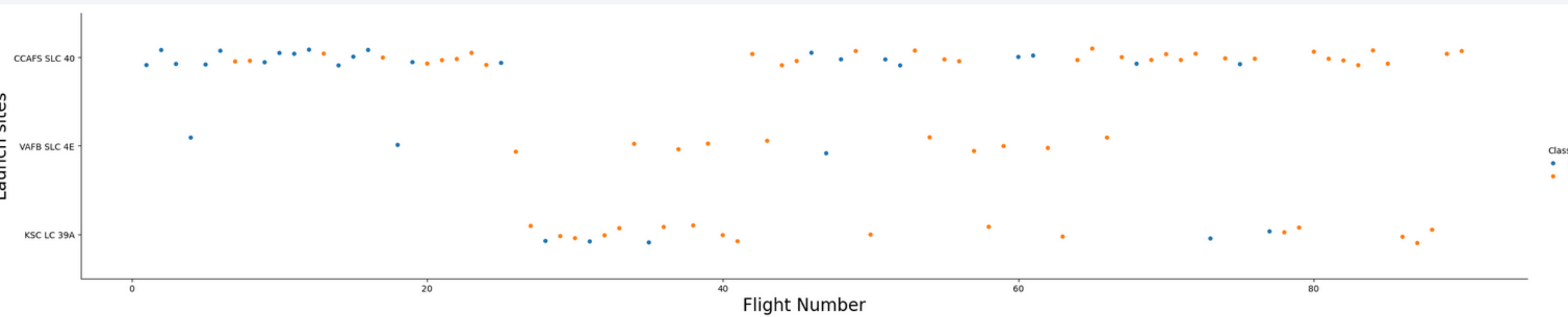
- The number of safe landings increases with time
- The launches happen mostly in CCAFS and, later, in KSC, in the east coast
- The higher the payload mass (especially in launch site CCAFS), the higher the success rate
- In terms of Orbit Type, GEO, HEO, SSO, VLEO had the most success rate.
- The orbit types changes with time (now more VLEO)

The background of the slide is an abstract composition. It features a dark blue field on the left side, which transitions into a complex pattern of diagonal streaks in shades of blue, red, and cyan on the right. These streaks have a textured, almost woven appearance. Overlaid on this pattern is a faint, light blue grid that recedes into the distance, creating a sense of depth and perspective.

Section 2

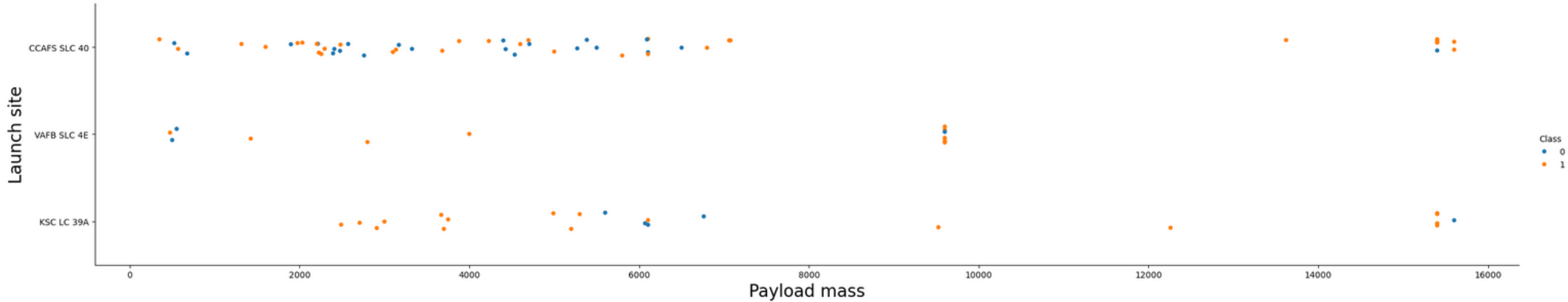
Insights drawn from EDA

Flight Number vs. Launch Site



- The number of safe landings increases with time
- The launches happen mostly in CCAFS and, later, in KSC, in the east coast

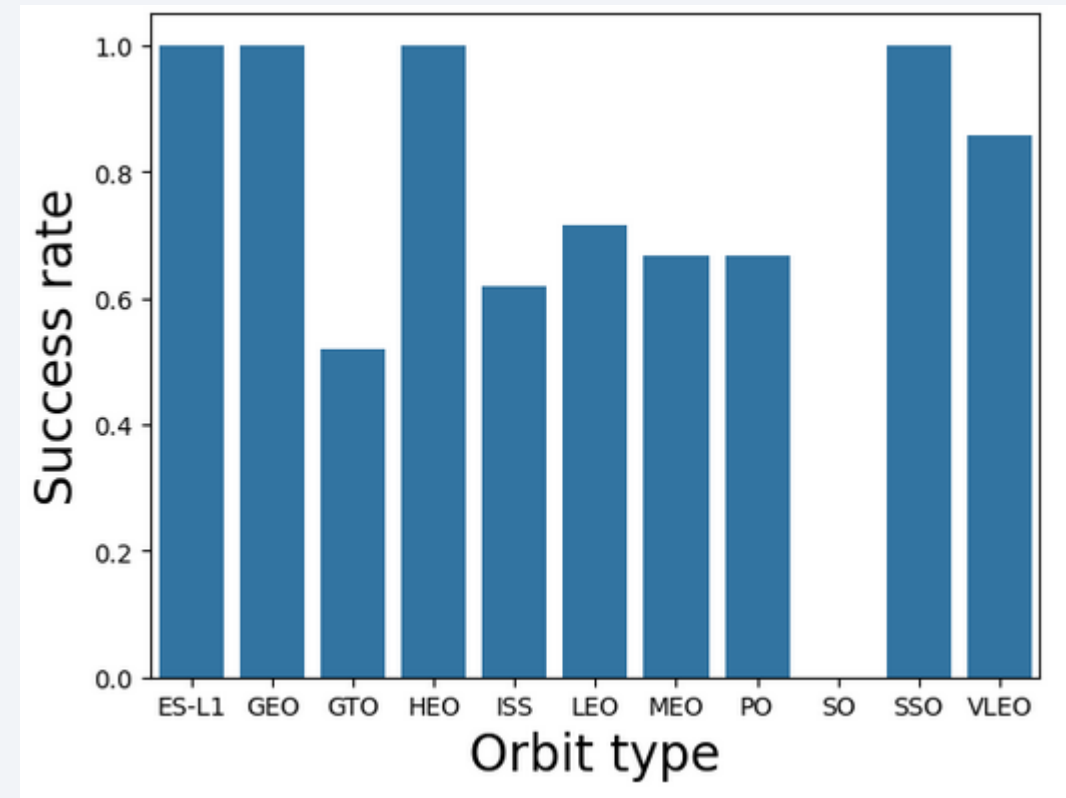
Payload vs. Launch Site



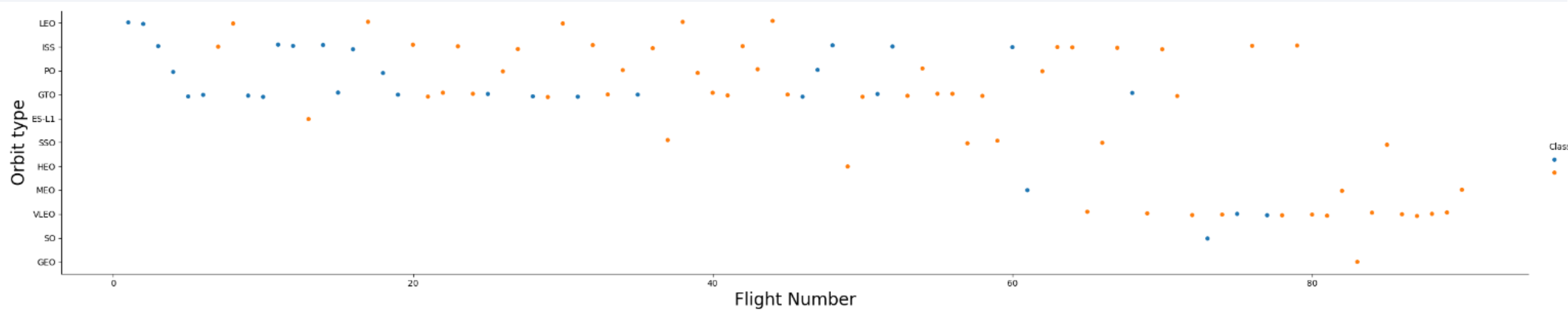
- The higher the payload mass (especially in launch site CCAFS), the higher the success rate

Success Rate vs. Orbit Type

- In terms of Orbit Type, GEO, HEO, SSO, VLEO had the most success rate.

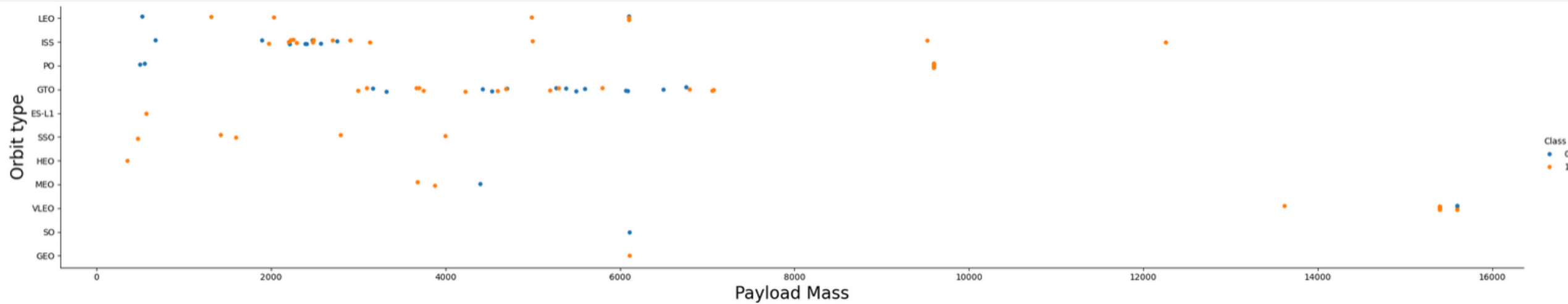


Flight Number vs. Orbit Type



- The orbit types changes with time (now more VLEO)

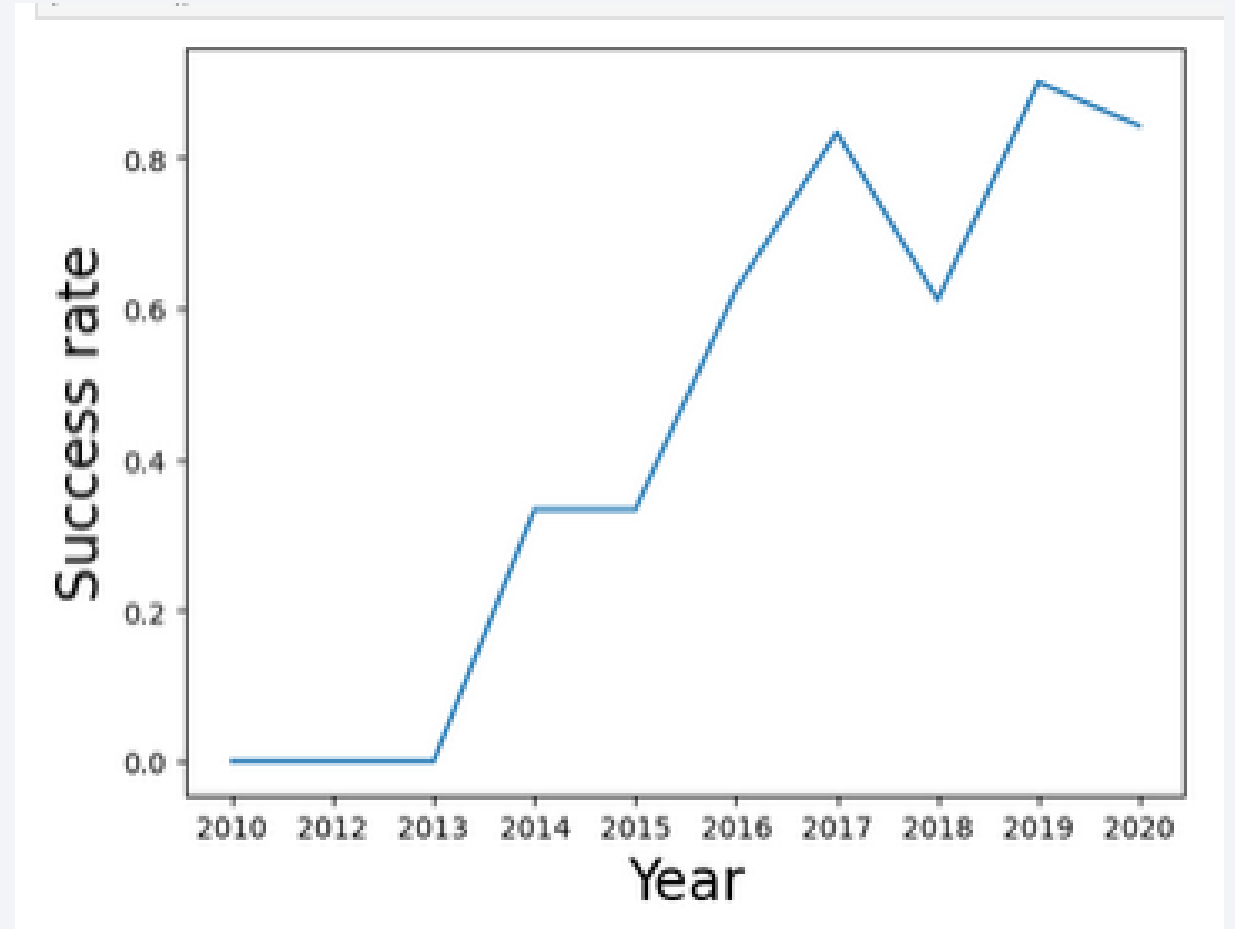
Payload vs. Orbit Type



- ISS: light payload
- VLEO: heavy payload
- GTO: in between

Launch Success Yearly Trend

- Success rate increases with time



All Launch Site Names

4 launch sites

Launch Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Launch Site Names Begin with 'CCA'

First five launches from CCAFS LC-40

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

45596kg carried in total for NASA

SUM(PAYLOAD_MASS_KG_)

45596

Average Payload Mass by F9 v1.1

2535 carried on average by booster version F9 v1.1

```
*****  
AVG(PAYLOAD_MASS_KG_)  
-----  
2534.6666666666665
```

First Successful Ground Landing Date

First successful ground landing in 2015 dec 22

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2015-12-22	1:29:00	F9 FT B1019	CCAFS LC-40	OG2 Mission 2 11 Orbcomm-OG2 satellites	2034	LEO	Orbcomm	Success	Success (ground pad)

Successful Drone Ship Landing with Payload between 4000 and 6000

4 booster version who sucessfully land with a payload between 4000kg and 6000kg

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- 100 successful missions
- 1 failed mission

```
34]: %%sql
      SELECT COUNT(*) FROM SPACEXTABLE WHERE Mission_Outcome LIKE "Success%"
      * sqlite:///my_data1.db

      Done.
34]: .....
      COUNT(*)
      -----
             100

35]: %%sql SELECT COUNT(*) FROM SPACEXTABLE WHERE Mission_Outcome LIKE "Failure%"
      * sqlite:///my_data1.db

      Done.
35]: .....
      COUNT(*)
      -----
             1
```

Boosters Carried Maximum Payload

12 booster versions carried the maximum payload
(15600kg)

Booster_Version

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

A lot of drone ship landing attempts between the date 2010-06-04 and 2017-03-20

Landing_Outcome	COUNT(Landing_Outcome)
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

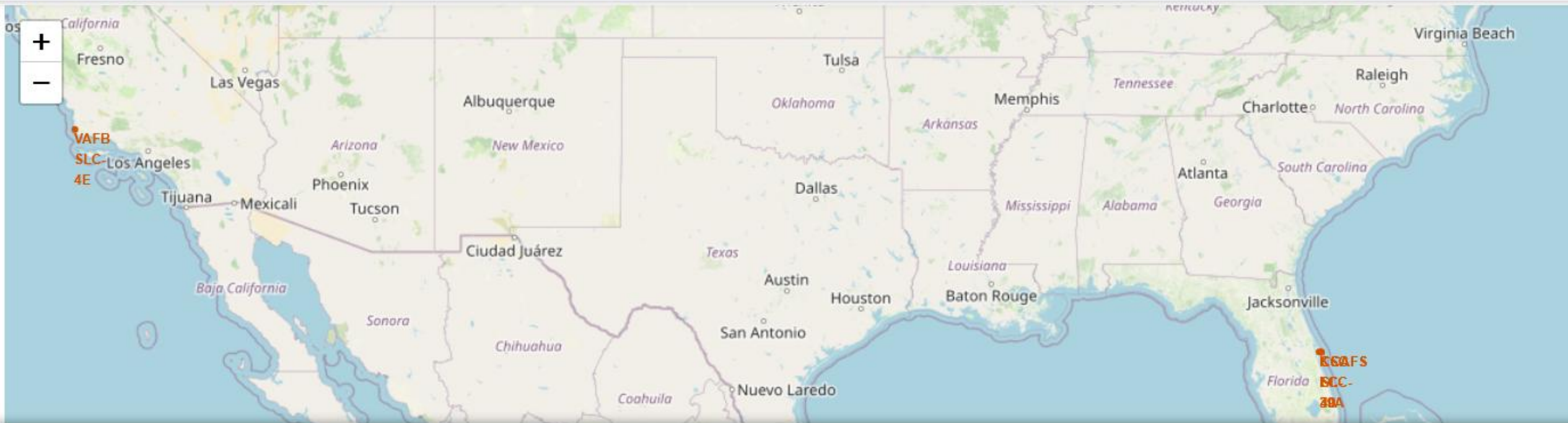
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky and a view of the Earth's surface, which is covered in a dense network of city lights and clouds. The lights are concentrated in the lower right portion of the image, while the upper left portion shows a clear blue sky.

Section 3

Launch Sites Proximities Analysis

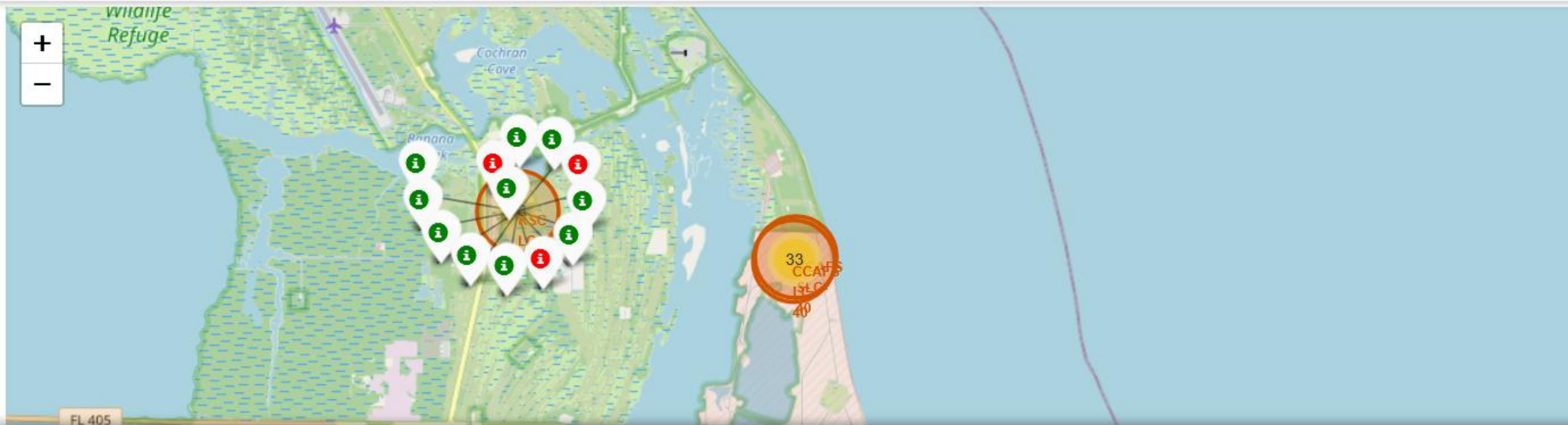
Launch sites locations

All launch sites locations



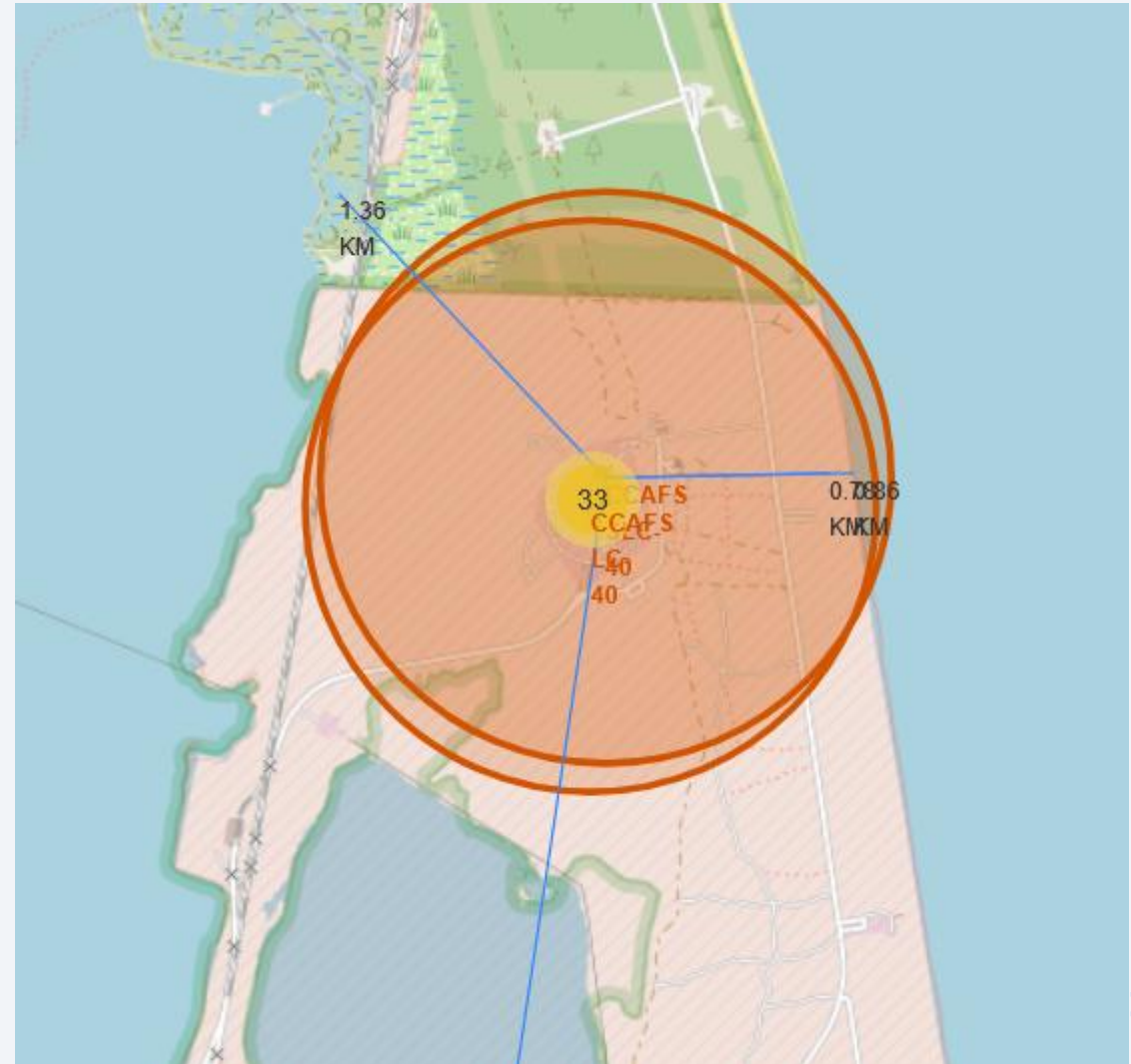
Landing attempts in KSC LC-39A

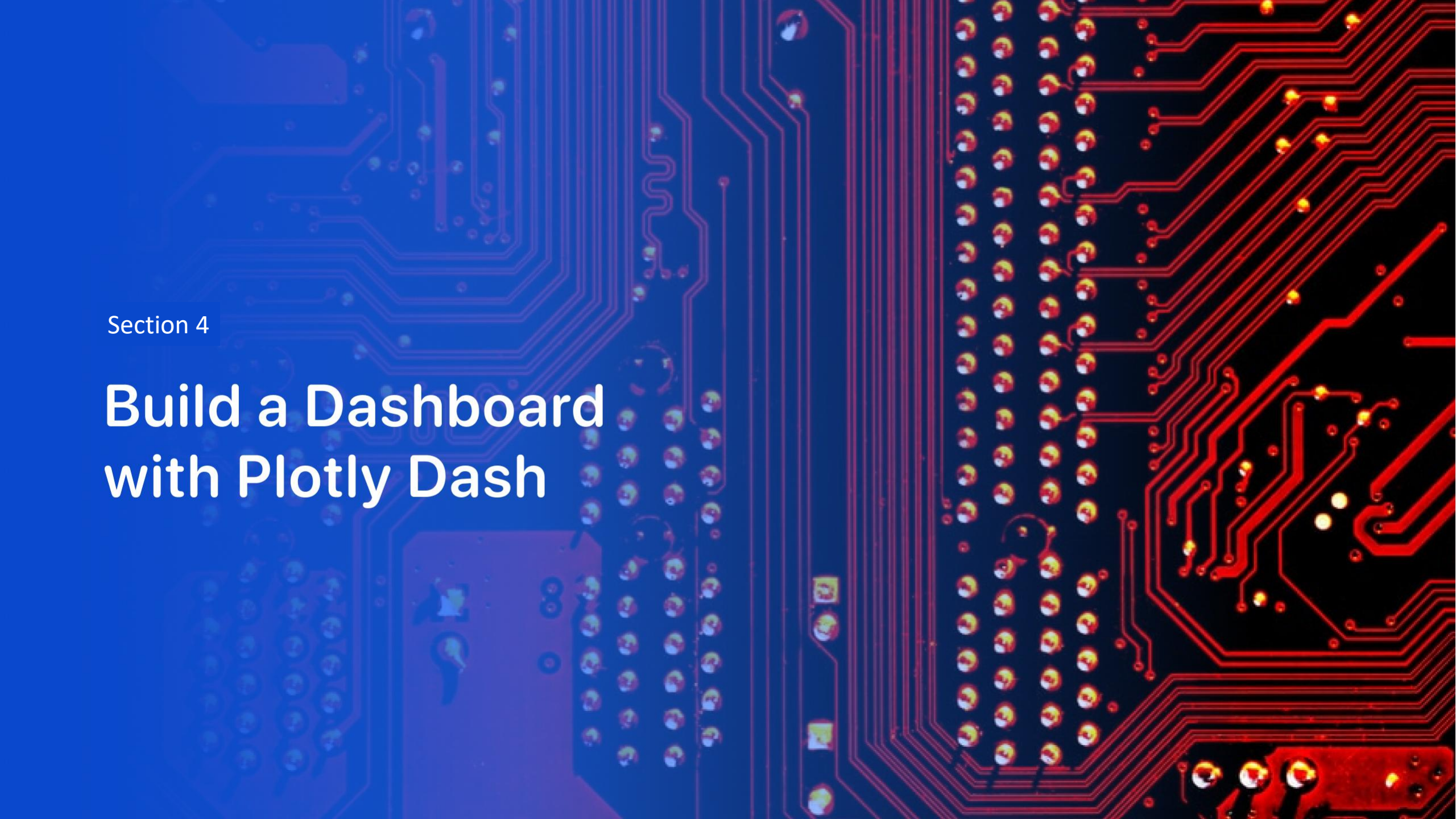
Green successful landing, red failed landing



Proximity of CCAFS to coast line, railway and city

The distance to the nearest city (Melbourne, FL) is ~ 51km.





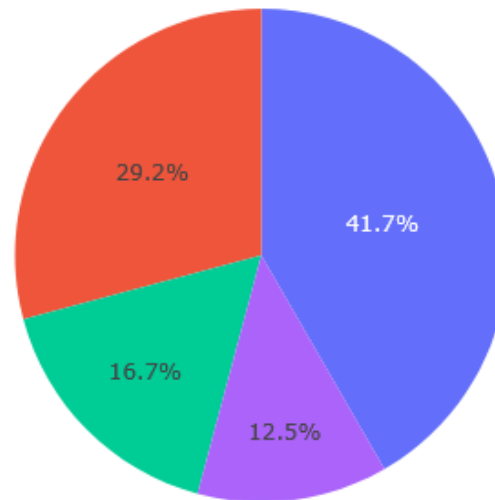
Section 4

Build a Dashboard with Plotly Dash

Success count for all launch sites

Most success landing in KSC LC-39A

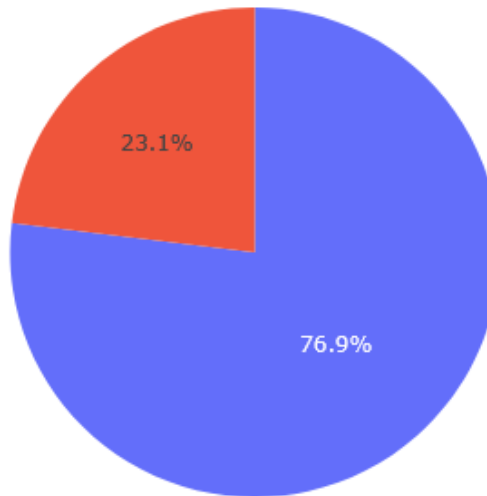
Success Count for all launch sites



Total success launches for site KSC LC-39A

Highest launch success ratio with 76.9%

Total Success Launches for site KSC LC-39A



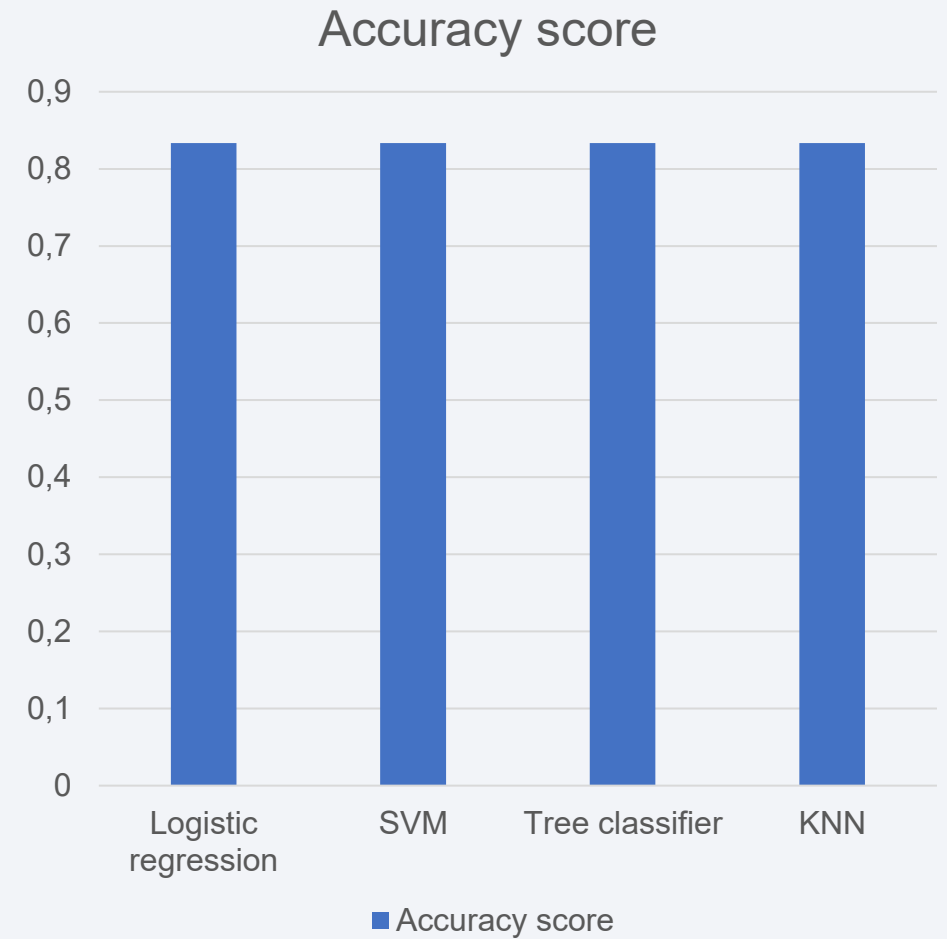


Section 5

Predictive Analysis (Classification)

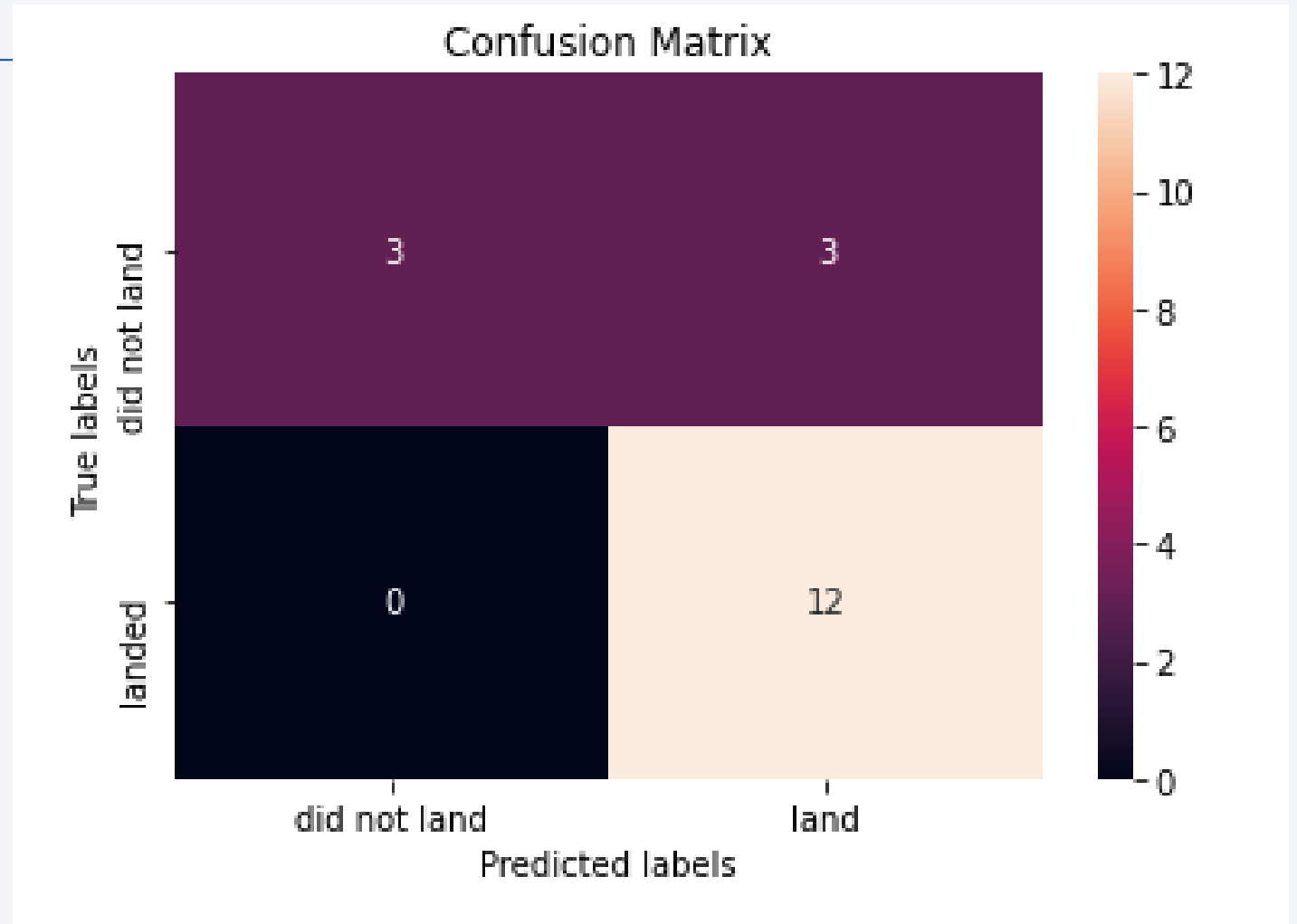
Classification Accuracy

- All methods show the same accuracy



Confusion Matrix

- All methods show the same confusion matrix



Conclusions

- The number of safe landings increases with time
- The launches happen mostly in CCAFS and, later, in KSC, in the east coast
- The higher the payload mass (especially in launch site CCAFS), the higher the success rate
- In terms of Orbit Type, GEO, HEO, SSO, VLEO had the most success rate.
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Thank you!

